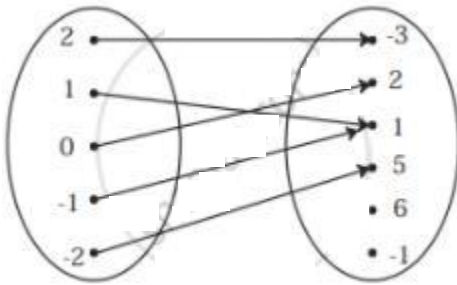


Chapter 2: Functions

EXERCISE 2.1 [PAGES 30 - 31]

Exercise 2.1 | Q 1.1 | Page 30

Check if the following relation is function.



SOLUTION

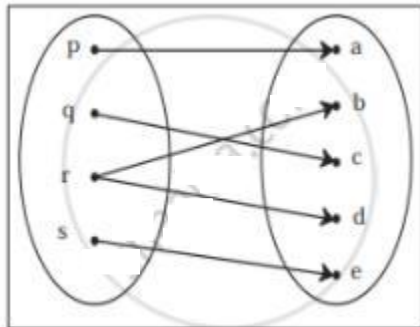
Yes

Reason:

Every element of set A has been assigned a unique element in set B.

Exercise 2.1 | Q 1.2 | Page 31

Check if the following relation is function.



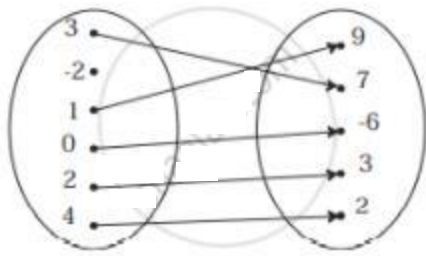
No.

Reason:

An element of set A has been assigned more than one element from set B.

Exercise 2.1 | Q 1.3 | Page 31

Check if the following relation is function.



SOLUTION

No.

Reason:

Not every element of set A has been assigned an image from set B.

Exercise 2.1 | Q 2.1 | Page 31

Which sets of ordered pairs represent functions from $A = \{1, 2, 3, 4\}$ to $B = \{-1, 0, 1, 2, 3\}$? Justify

$\{(1, 0), (3, 3), (2, -1), (4, 1), (2, 2)\}$

SOLUTION

$\{(1, 0), (3, 3), (2, -1), (4, 1), (2, 2)\}$ does not represent a function.

Reason:

$(2, -1)$ and $(2, 2)$ show that element $2 \in A$ has been assigned two images – 1 and 2 from set B.

Exercise 2.1 | Q 2.2 | Page 31

Which sets of ordered pairs represent functions from $A = \{1, 2, 3, 4\}$ to $B = \{-1, 0, 1, 2, 3\}$? Justify

$\{(1, 2), (2, -1), (3, 1), (4, 3)\}$

SOLUTION

$\{(1, 2), (2, -1), (3, 1), (4, 3)\}$ represents a function. **Reason:** Every element of set A has a unique image in set B.

Exercise 2.1 | Q 2.3 | Page 31

Which set of ordered pair represent function from $A = \{1, 2, 3, 4\}$ to $B = \{-1, 0, 1, 2, 3\}$? Justify.

$\{(1, 3), (4, 1), (2, 2)\}$

SOLUTION

$\{(1, 3), (4, 1), (2, 2)\}$ does not represent a function.

Reason:

$3 \in A$ does not have an image in set B.

Exercise 2.1 | Q 2.4 | Page 31

Which set of ordered pair represent function from $A = \{1,2,3,4\}$ to $B = \{-1,0,1,2,3\}$? Justify
 $\{(1,1), (2,1), (3,1), (4,1)\}$

SOLUTION

$\{(1, 1), (2, 1), (3, 1), (4, 1)\}$ represents a function **Reason:**
 Every element of set A has been assigned a unique image in set B.

Exercise 2.1 | Q 3.1 | Page 31

If $f(m) = m^2 - 3m + 1$, find $f(0)$

SOLUTION

$$f(m) = m^2 - 3m + 1$$

$$f(0) = 0^2 - 3(0) + 1 = 1$$

Exercise 2.1 | Q 3.2 | Page 31

If $f(m) = m^2 - 3m + 1$, find $f(-3)$

SOLUTION

$$f(-3) = (-3)^2 - 3(-3) + 1$$

$$= 9 + 9 + 1 = 19$$

Exercise 2.1 | Q 3.3 | Page 31

If $f(m) = m^2 - 3m + 1$, find $f\left(\frac{1}{2}\right)$

SOLUTION

$$f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 - 3\left(\frac{1}{2}\right) + 1 = \frac{1}{4} - \frac{3}{2} + 1$$

$$= \frac{1 - 6 + 4}{4} = -\frac{1}{4}$$

Exercise 2.1 | Q 3.4 | Page 31

If $f(m) = m^2 - 3m + 1$, find $f(x + 1)$

SOLUTION

$$f(x + 1) = (x + 1)^2 - 3(x + 1) + 1$$

$$= x^2 + 2x + 1 - 3x - 3 + 1$$

$$= x^2 - x - 1$$

Exercise 2.1 | Q 3.5 | Page 31

If $f(m) = m^2 - 3m + 1$, find $f(-x)$

SOLUTION

$$\begin{aligned}f(-x) &= (-x)^2 - 3(-x) + 1 \\&= x^2 + 3x + 1\end{aligned}$$

Exercise 2.1 | Q 4.1 | Page 31

Find x , if $g(x) = 0$ where

$$g(x) = \frac{5x - 6}{7}$$

SOLUTION

$$g(x) = \frac{5x - 6}{7}$$

$$g(x) = 0$$

$$\therefore \frac{5x - 6}{7} = 0$$

$$\therefore 5x - 6 = 0$$

$$\therefore x = \frac{6}{5}$$

Exercise 2.1 | Q 4.2 | Page 31

Find x , if $g(x) = 0$ where

$$g(x) = \frac{18 - 2x^2}{7}$$

SOLUTION

$$g(x) = \frac{18 - 2x^2}{7}$$

$$g(x) = 0$$

$$\therefore \frac{18 - 2x^2}{7} = 0$$

$$\therefore 18 - 2x^2 = 0$$

$$\therefore x^2 = \frac{18}{2} = 9$$

$$\therefore x = \pm 3$$

Exercise 2.1 | Q 4.3 | Page 31

Find x, if $g(x) = 0$ where

$$g(x) = 6x^2 + x - 2$$

SOLUTION

$$g(x) = 6x^2 + x - 2$$

$$g(x) = 0$$

$$\therefore 6x^2 + x - 2 = 0$$

$$\therefore 6x^2 + 4x - 3x - 2 = 0$$

$$\therefore 2x(3x + 2) - 1(3x + 2) = 0$$

$$\therefore (2x - 1)(3x + 2) = 0$$

$$\therefore 2x - 1 = 0 \text{ or } 3x + 2 = 0$$

$$\therefore x = \frac{1}{2} \text{ or } x = -\frac{2}{3}$$

Exercise 2.1 | Q 5 | Page 31

Find x, if $f(x) = g(x)$ where

$$f(x) = x^4 + 2x^2, g(x) = 11x^2$$

SOLUTION

$$f(x) = x^4 + 2x^2, g(x) = 11x^2$$

$$f(x) = g(x)$$

$$\therefore x^4 + 2x^2 = 11x^2$$

$$\therefore x^4 - 9x^2 = 0$$

$$\therefore x^2(x^2 - 9) = 0$$

$$\therefore x = 0 \text{ or } x^2 - 9 = 0$$

$$\therefore x = 0 \text{ or } x^2 = 9$$

$$\therefore x = 0 \text{ or } x = \pm 3$$

Exercise 2.1 | Q 6 | Page 31

If $(x) = \{x^2 + 3, x \leq 2, 5x + 7, x > 2\}$, then find

$$f(3)$$

$$f(2)$$

$$f(0)$$

SOLUTION

$$x^2 + 3, x \leq 2, 5x + 7, x > 2$$

$$\text{i. } f(3) = 5(3) + 7 = 15 + 7 = 22$$

$$\text{ii. } f(2) = 2^2 + 3 = 4 + 3 = 7$$

$$\text{iii. } f(0) = 0^2 + 3 = 3$$

Exercise 2.1 | Q 7 | Page 31

$$\text{If } f(x) = \begin{cases} 4x - 2, \\ \end{cases}$$

$$x \leq -3$$

$$5, -3 < x < 3,$$

$$x^2,$$

$$x \geq 3 \text{ then find } f(-4), f(-3), f(1), f(5)$$

SOLUTION

$$f(x) = 4x - 2,$$

$$x \leq -3$$

$$5, -3 < x < 3,$$

$$x^2,$$

$$x \geq 3$$

$$\text{i. } f(-4) = 4(-4) - 2 = -16 - 2 = -18$$

$$\text{ii. } f(-3) = 4(-3) - 2 = -12 - 2 = -14$$

$$\text{iii. } f(1) = 5$$

$$\text{iv. } f(5) = 5^2 = 25$$

Exercise 2.1 | Q 8.1 | Page 31

$$\text{If } f(x) = 3x + 5, g(x) = 6x - 1, \text{ then find } (f+g)(x)$$

SOLUTION

$$f(x) = 3x + 5, g(x) = 6x - 1$$

$$(f+g)(x) = f(x) + g(x)$$

$$= 3x + 5 + 6x - 1 = 9x + 4$$

Exercise 2.1 | Q 8.2 | Page 31

$$\text{If } f(x) = 3x + 5, g(x) = 6x - 1, \text{ then find } (f-g)(2)$$

SOLUTION

$$(f-g)(2) = f(2) - g(2)$$

$$= [3(2) + 5] - [6(2) - 1]$$

$$= 6 + 5 - 12 + 1 = 0$$

Exercise 2.1 | Q 8.3 | Page 31

If $f(x) = 3x + 5$, $g(x) = 6x - 1$, then find $(f \circ g)(3)$

SOLUTION

$$(f \circ g)(3) = f(g(3))$$

$$= [3(3) + 5] [6(3) - 1]$$

$$= (14)(17) = 238$$

Exercise 2.1 | Q 8.4 | Page 31

If $f(x) = 3x + 5$, $g(x) = 6x - 1$, then find $\left(\frac{f}{g}\right)(x)$ and its domain

SOLUTION

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{3x + 5}{6x - 1}, x \neq \frac{1}{6}$$

$$\text{Domain} = \mathbb{R} - \left\{\frac{1}{6}\right\}$$

Exercise 2.1 | Q 9.1 | Page 31

If $f(x) = 2x^2 + 3$, $g(x) = 5x - 2$, then find $f \circ g$

SOLUTION

$$f(x) = 2x^2 + 3, g(x) = 5x - 2$$

$$(f \circ g)(x) = f(g(x)) = f(5x - 2)$$

$$= 2(5x - 2)^2 + 3$$

$$= 2(25x^2 - 20x + 4) + 3$$

$$= 50x^2 - 40x + 8 + 3$$

$$= 50x^2 - 40x + 11$$

Exercise 2.1 | Q 9.2 | Page 31

If $f(x) = 2x^2 + 3$, $g(x) = 5x - 2$, then find $g \circ f$

SOLUTION

$$(g \circ f)(x) = g(f(x)) = g(2x^2 + 3)$$

$$= 5(2x^2 + 3) - 2$$

$$= 10x^2 + 15 - 2$$

$$= 10x^2 + 13$$

NOTES

$$(g \circ f)(x) = g(f(x)) = g(2x^2 + 3)$$

$$= 5(2x^2 + 3) - 2$$

$$= 10x^2 + 15 - 2$$

$$= 10x^2 + 13$$

Exercise 2.1 | Q 9.3 | Page 31

If $f(x) = 2x^2 + 3$, $g(x) = 5x - 2$, then find $f \circ g$

SOLUTION

$$(f \circ g)(x) = f(g(x)) = f(5x - 2)$$

$$= 2(5x - 2)^2 + 3$$

$$= 2(25x^2 - 20x + 4) + 3$$

$$= 50x^2 - 40x + 8 + 3$$

$$= 50x^2 - 40x + 11$$

Exercise 2.1 | Q 9.4 | Page 31

If $f(x) = 2x^2 + 3$, $g(x) = 5x - 2$, then find $g \circ g$

SOLUTION

$$(g \circ g)(x) = g(g(x)) = g(5x - 2)$$

$$= 5(5x - 2) - 2$$

$$= 25x - 10 - 2$$

$$= 25x - 12$$

MISCELLANEOUS EXERCISE 2 [PAGE 32]

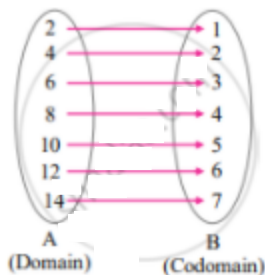
Miscellaneous Exercise 2 | Q 1.1 | Page 32

Which of the following relations are functions? If it is a function determine its domain and range.

$\{(2, 1), (4, 2), (6, 3), (8, 4), (10, 5), (12, 6), (14, 7)\}$

SOLUTION

$\{(2, 1), (4, 2), (6, 3), (8, 4), (10, 5), (12, 6), (14, 7)\}$



Every element of set A has been assigned a unique element in set B.

∴ Given relation is a function.

Domain = {2, 4, 6, 8, 10, 12, 14},

Range = {1, 2, 3, 4, 5, 6, 7}

Miscellaneous Exercise 2 | Q 1.2 | Page 32

Which of the following relations are functions? If it is a function determine its domain and range.

$\{(0, 0), (1, 1), (1, -1), (4, 2), (4, -2), (9, 3), (9, -3), (16, 4), (16, -4)\}$

SOLUTION

$\{(0, 0), (1, 1), (1, -1), (4, 2), (4, -2), (9, 3), (9, -3), (16, 4), (16, -4)\}$

∴ $(1, 1), (1, -1) \in$ the relation

∴ Given relation is not a function. As the element 1 of the domain has not been assigned a unique element of co-domain.

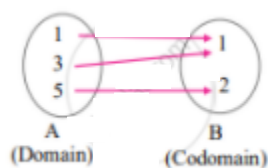
Miscellaneous Exercise 2 | Q 1.3 | Page 32

Which of the following relations are functions? If it is a function determine its domain and range.

$\{(1, 1), (3, 1), (5, 2)\}$

SOLUTION

$\{(1, 1), (3, 1), (5, 2)\}$



Every element of set A has been assigned a unique element in set B.

∴ Given relation is a function.

Domain = {1, 3, 5}, Range = {1, 2}

Miscellaneous Exercise 2 | Q 2 | Page 32

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3\frac{x}{5} + 2$, $x \in \mathbb{R}$. Show that f is one-one and onto. Hence find f^{-1}

SOLUTION

$f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{3x}{5} + 2$

First we have to prove that f is one-one function for that we have to prove if $f(x_1) = f(x_2)$ then $x_1 = x_2$

$$\text{Here } f(x) = \frac{3x}{5} + 2$$

$$\text{Let } f(x_1) = f(x_2)$$

$$\therefore \frac{3x_1}{5} + 2 = \frac{3x_2}{5} + 2$$

$$\therefore \frac{3x_1}{5} = \frac{3x_2}{5}$$

$$\therefore x_1 = x_2$$

$\therefore f$ is a one-one function. Now, we have to prove that f is an onto function. Let $y \in \mathbb{R}$ be such that $y = f(x)$

$$\therefore y = \frac{3x}{5} + 2$$

$$\therefore y - 2 = \frac{3x}{5}$$

$$\therefore x = \frac{5(y - 2)}{3} \in \mathbb{R}$$

$\therefore$ for any $y \in \text{co-domain } \mathbb{R}$, there exist an element $x = \frac{5(y - 2)}{3}$ domain $\mathbb{R}$ such that $f(x) = y$

$\therefore f$ is an onto function.

$\therefore f$ is one-one onto function.

$\therefore f^{-1}$ exists

$$\therefore f^{-1}(y) = \frac{5(y-2)}{3}$$

$$\therefore f^{-1}(x) = \frac{5(x-2)}{3}$$

Miscellaneous Exercise 2 | Q 3 | Page 32

A function f is defined as follows

$$f(x) = 4x + 5, \text{ for } -4 \leq x < 0.$$

Find the values of $f(-1)$, $f(-2)$, $f(0)$, if they exist.

SOLUTION

$$f(x) = 4x + 5, -4 \leq x < 0$$

$$f(-1) = 4(-1) + 5 = -4 + 5 = 1$$

$$f(-2) = 4(-2) + 5 = -8 + 5 = -3$$

$x = 0 \notin \text{domain of } f$

$\therefore f(0)$ does not exist.

Miscellaneous Exercise 2 | Q 4 | Page 32

A function f is defined as follows: $f(x) = 5 - x$ for $0 \leq x \leq 4$ Find the value of x such that $f(x) = 3$

SOLUTION

$$f(x) = 5 - x$$

$$f(x) = 3$$

$$\therefore 5 - x = 3$$

$$\therefore x = 5 - 3 = 2$$

Miscellaneous Exercise 2 | Q 5 | Page 32

If $f(x) = 3x^2 - 5x + 7$ find $f(x-1)$.

SOLUTION

$$f(x) = 3x^2 - 5x + 7$$

$$\therefore f(x-1) = 3(x-1)^2 - 5(x-1) + 7$$

$$= 3(x^2 - 2x + 1) - 5(x-1) + 7$$

$$= 3x^2 - 6x + 3 - 5x + 5 + 7$$

$$= 3x^2 - 11x + 15$$

Miscellaneous Exercise 2 | Q 6 | Page 32

If $f(x) = 3x + a$ and $f(1) = 7$ find a and $f(4)$.

SOLUTION

$$f(x) = 3x + a$$

$$f(1) = 7$$

$$\therefore 3(1) + a = 7$$

$$\therefore a = 7 - 3 = 4$$

$$\therefore f(x) = 3x + 4$$

$$\therefore f(4) = 3(4) + 4 = 12 + 4 = 16$$

Miscellaneous Exercise 2 | Q 7 | Page 32

If $f(x) = ax^2 + bx + 2$ and $f(1) = 3$, $f(4) = 42$. find a and b .

SOLUTION

$$f(x) = ax^2 + bx + 2$$

$$f(1) = 3$$

$$\therefore a(1)^2 + b(1) + 2 = 3$$

$$\therefore a + b = 1 \dots (i)$$

$$\therefore f(4) = 42$$

$$\therefore a(4)^2 + b(4) + 2 = 42$$

$$\therefore 16a + 4b = 40$$

Dividing by 4, we get

$$4a + b = 10 \dots (ii)$$

Solving (i) and (ii), we get

$$a = 3, b = -2$$

Miscellaneous Exercise 2 | Q 8 | Page 32

$$\text{If } f(x) = \frac{2x - 1}{5x - 2}, x \neq \frac{2}{5}$$

Verify whether $(f \circ f)(x) = x$

SOLUTION

$$\begin{aligned}
 (f \circ f)(x) &= f(f(x)) \\
 &= f\left(\frac{2x-1}{5x-2}\right) \\
 &= \frac{2\frac{2x-1}{5x-2} - 1}{5\frac{2x-1}{5x-2} - 2} \\
 &= \frac{4x-2-5x+2}{10x-5-10x+4} = \frac{-x}{-1} = x
 \end{aligned}$$

Miscellaneous Exercise 2 | Q 9 | Page 32

If $f(x) = \frac{x+3}{4x-5}$, $g(x) = \frac{3+5x}{4x-1}$ then verify that $(f \circ g)(x) = x$.

SOLUTION

$$\begin{aligned}
 f(x) &= \frac{x+3}{4x-5}, \quad g(x) = \frac{3+5x}{4x-1} \\
 (f \circ g)(x) &= f(g(x)) \\
 &= f\left(\frac{3+5x}{4x-1}\right) \\
 &= \frac{\frac{3+5x}{4x-1} + 3}{4\frac{3+5x}{4x-1} - 5} \\
 &= \frac{3+5x+12x-3}{12+20x-20x+5} = \frac{17x}{17} = x
 \end{aligned}$$